H.T.No: | | | | | | | | | | | CourseCode:**241MA002**

# ADITYA UNIVERSITY

B.Tech – II Semester End Examinations Regular 2025

**DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS**
**(Common to all except Ag.E)**

**Time: 3 hours** **Max. Marks: 100**

**Answer ONE question from each unit**
**All Questions Carry Equal Marks**
**All parts of the questions must be answered at one place only**

MODEL QUESTION PAPER

**UNIT-I**

| | | | | | |
|---|---|---|---|---|---|
| 1 | a. | Solve the differential equation (2xlnx-xy)dy+2ydx=0. | L3 | CO1 | [10M] |
| | b. | Obtain the solution of $(1+x^2)\frac{dy}{dx}$+y=$e^{tan^{-1}x}$. | L3 | CO1 | [10M] |

**(OR)**

| | | | | | |
|---|---|---|---|---|---|
| 2 | a. | Find orthogonal trajectories of family of circles $x^2+y^2=ax$. | L3 | CO1 | [10M] |
| | b. | A generator having emf 100 volts is connected in series with a 10 ohm resistor and an inductance of 2 Henries. If the switch is closed at time t=0 then find the current at time t>0. | L3 | CO1 | [10M] |

**UNIT-II**

| | | | | | |
|---|---|---|---|---|---|
| 3 | a. | Solve $(D^2+4)y=e^{3x}+sin2x$. | L3 | CO2 | [10M] |
| | b. | Using method of variation of parameters, find the solution of the differential equation $\frac{d^2y}{dx^2}$-6$\frac{dy}{dx}$+9=$\frac{e^{3x}}{x^2}$. | L3 | CO2 | [10M] |

**(OR)**

| | | | | | |
|---|---|---|---|---|---|
| 4 | a. | Solve $x^2\frac{d^2y}{dx^2}+4x\frac{dy}{dx}+2y=e^x$. | L3 | CO2 | [10M] |
| | b. | Explain about modelling of LCR circuit . | L2 | CO2 | [10M] |

**UNIT-III**

| | | | | | |
|---|---|---|---|---|---|
| 5 | a. | Solve $x^2(y-z)p+y^2(z-x)q=z^2(x-y)$. | L3 | CO3 | [10M] |
| | b. | Solve $\frac{\partial^4 z}{\partial x^4}-\frac{\partial^4 z}{\partial y^4}=0$. | L3 | CO3 | [10M] |

**(OR)**

| | | | | | |
|---|---|---|---|---|---|
| 6 | a. | Solve $\frac{\partial^3 z}{\partial x^3}-3\frac{\partial^3 z}{\partial x^2\partial y}+4\frac{\partial^3 z}{\partial y^3}=e^{x+2y}$. | L3 | CO3 | [10M] |
| | b. | Solve the partial differential equation z(x-y)=px$^2$-qy$^2$ | L3 | CO3 | [10M] |

**UNIT-IV**

| | | | | | |
|---|---|---|---|---|---|
| 7 | a. | If u=x+y+z, v=x$^2$+y$^2$+z$^2$, w=xy+yz+zx then find [grad u grad v grad w] | L3 | CO4 | [10M] |
| | b. | Find directional derivative of 4xy$^2$+2x$^2$yz at a point A(1,2,3) in the direction of line AB where B=(5,0,4). | L3 | CO4 | [10M] |

**(OR)**

| | | | | | |
|---|---|---|---|---|---|
| 8 | a. | Find divergence of F where F=grad($x^3+y^3+z^3-3xyz$). | L3 | CO5 | [8M] |
| | b. | Show that $(y^2-z^2+3yz-zx)i+(3xz+2xy)j+(3xy-2xz+2z)k$ is irrotational and hence find its scalar potential. | L3 | CO5 | [12M] |

**UNIT-V**

| | | | | | |
|---|---|---|---|---|---|
| 9 | a. | Find work done in moving a particle in the force field $F=3x^2i+j+zk$ along the straight line from (0,0,0) to (2,1,3). | L3 | CO5 | [10M] |
| | b. | Evaluate $\int_S F.n\,ds$ where F=18zi-12j+3yk and S is the surface of the plane 2x+3y+6z=12 located in the first octant. | L3 | CO5 | [10M] |

**(OR)**

| | | | | | |
|---|---|---|---|---|---|
| 10 | a. | Verify Green's theorem in a plane $\oint_C (xy^2 - 2xy)dx + (x^2y + 3)dy$ around the boundary C of the region bounded by $y^2=8x$ and x=2. | L3 | CO5 | [14M] |
| | b. | State (i) Gauss divergence theorem (ii) Stoke's theorem. | L1 | CO5 | [6M] |

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MODEL QUESTION PAPER